HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · INFRASTRUCTURE & CONSTRUCTION · PROJECTS & OPERATIONS

EPC CONTRACTOR · HIGHWAYS, EXPRESSWAYS AND RAIL · MANY SITES

Project Control Room

From site report to boardroom, every kilometre in view. GenAI reads daily reports, voice notes and site photos; progress is tracked by chainage; fleet and fuel are watched machine by machine; tenders are read in hours; and safety is checked from photos.

~~52%~~ → **71%** fleet utilisation across sites



6 weeks
Schedule slippage flagged earlier
from the daily report, not the monthly review



71 %
Fleet utilisation
▲ from about 52%



4 %
Fuel variance against norms
▼ from about 14%

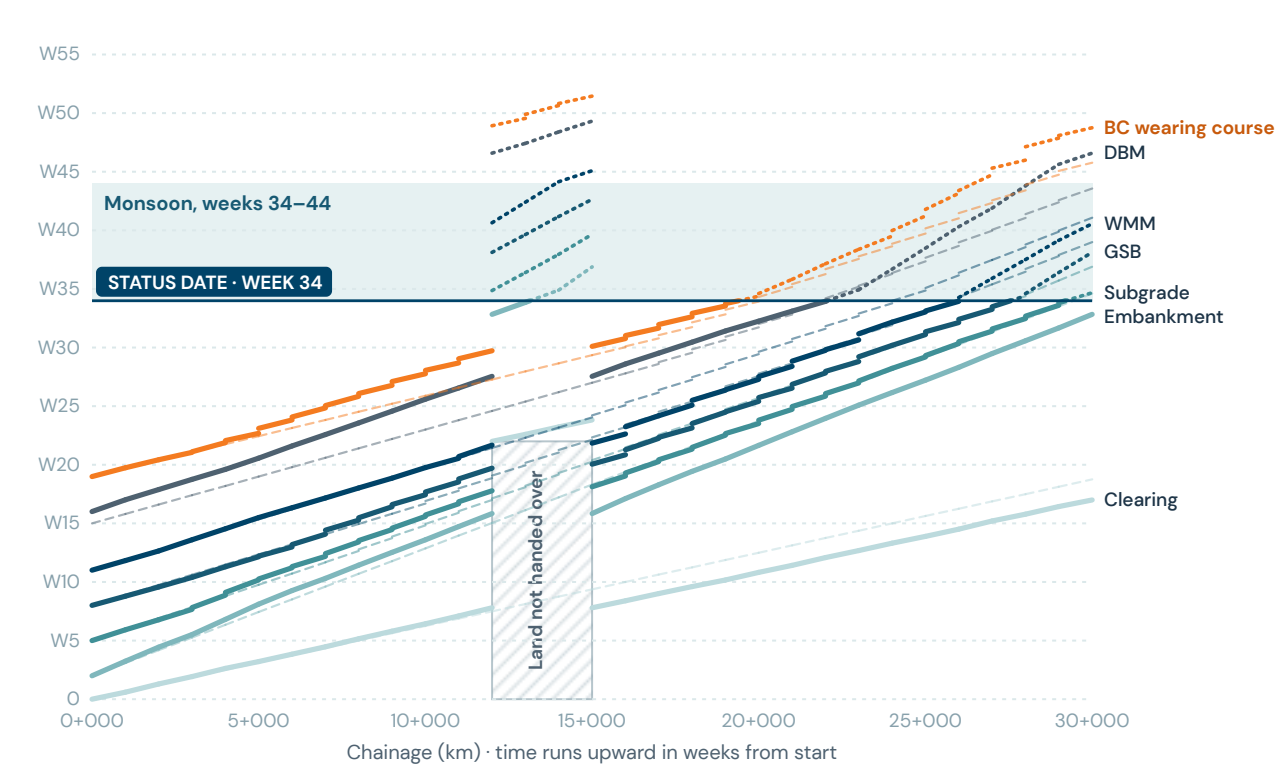


2 hours
First read of a tender document
▼ from about 3 days

Rings: one segment per week gained; utilisation with the earlier 52% inside; fuel variance (4%) with the earlier 14% inside, on a 0–100% scale; 2 hours as a share of 3 days (72 h). Figures indicative; validated metrics available on request.

01 Planned versus actual, kilometre by kilometre

Time-chainage view of one 30 km road project at week 34



Planned · Actual to status date · Forecast

Each line is one pavement layer moving along the road; **orange** = the layer furthest behind

PLANNED
84.3%
by status date

ACTUAL
77.4%
weighted by cost

SPI
0.92
slipping

FORECAST DELAY
+86 d
finish Dec 2026 vs plan Sept 2026

MOST BEHIND PLAN, BY LAYER (POINTS)

12.0 BC **9.8** DBM **4.7** Subgrade **4.5** GSB **4.4** Embankment **3.3** WMM

A WHATSAPP-STYLE REPORT, READ BY GENAI

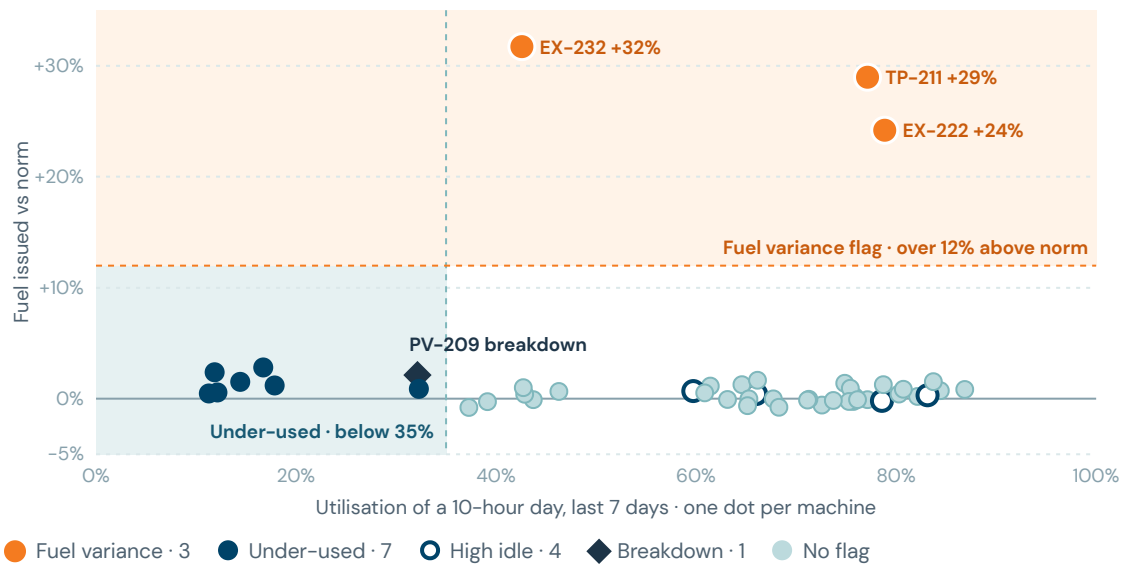
"Ch 17+400 to 18+150 DBM laid 750 m with 2 pavers. Ch 21+000 to 21+800 WMM 800 m completed. Ch 6+200 to 7+000 BC 800 m."

ACTIVITY	CHAINAGE	LENGTH	AGAINST PLAN
DBM	17+400 – 18+150	750 m	4.4 wk behind
WMM	21+000 – 21+800	800 m	2.4 wk behind
BC	6+200 – 7+000	800 m	10.2 wk behind

ILLUSTRATIVE · synthetic project, chainages and schedule from the demo; activity weights and rates are illustrative.

02 Every machine: working, idle or leaking diesel

7 days of telematics · 48 machines · 7 locations



Fuel variance · 3 · Under-used · 7 · High idle · 4 · Breakdown · 1 · No flag

48	60%	21%	2.3%	₹2.4 L
machines	utilisation of 10 h	idle share of engine hours	fuel issued vs norm	excess diesel a month

REDEPLOY BEFORE YOU HIRE

Paver Plant yard → Site A — **2 days**
DBM behind plan and a paver down · 156 km · used 12% now — payback · ₹9 L/month hire avoided

Motor grader Plant yard → Site A — **4 days**
Subgrade crew short of a grader · 156 km · used 12% now — payback · ₹4.5 L/month hire avoided

2 excavators Site E → Site C — **17 days**
Embankment at about 80% of planned rate · 734 km · used 14–17% now — payback · ₹3.2 L/month each

ILLUSTRATIVE · synthetic fleet. Fuel norms in litres per working hour by machine type; idle burn at 35% of the norm; diesel at ₹92 a litre.

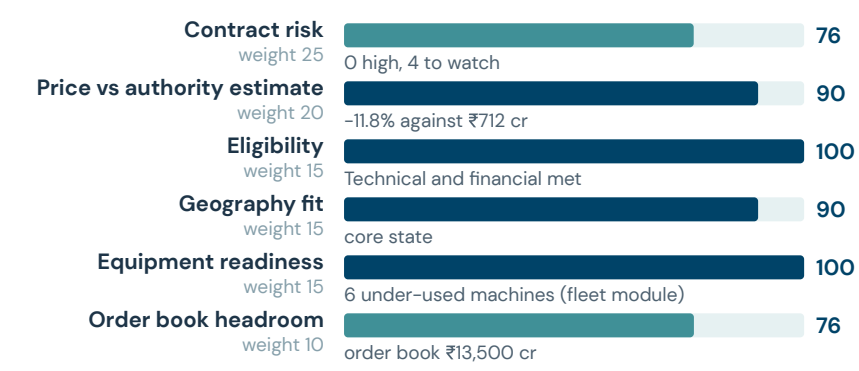
03 Tenders read in hours, sites scored from photos

Two more lenses of the same control room

Bid / no-bid score for a sample highway RFP

Clauses pulled out and rated; bill of quantities priced from the rate library

88 of 100 · bid at ₹628 cr, 11.8% under the ₹712 cr authority estimate

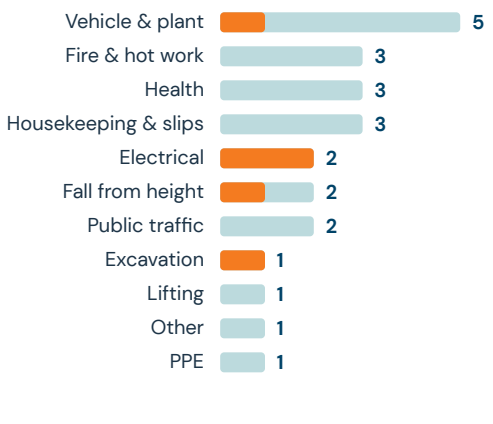


Length **30 km** · Build **730 days** · Land at start **90%** · LDs **0.05%/day, cap 10%** · O&M **15 years** · Payment in build **40% of BPC** · Defects liability **not found · check manually**

ILLUSTRATIVE · sample RFPs are synthetic extracts; rate library, net worth and order book are illustrative.

Incidents sorted, every site scored

24 incident and near-miss notes, last 90 days · **orange** = high potential (5)



SITE	OBS. H·M·L	HIGH-POT.	TRAINED	SCORE
Site C	3 · 6 · 5	2	79%	53
Site A	2 · 5 · 6	1	88%	72
Site E	1 · 4 · 6	1	84%	76
Site B	1 · 3 · 4	1	93%	85
Site F	0 · 1 · 2	0	97%	100
Site D	0 · 2 · 3	0	95%	100
Plant yard	0 · 1 · 1	0	90%	100

Photo check: **No helmet · High** → stop work until a helmet is worn; toolbox talk for the crew. **No hi-vis vest · Medium** → issue a vest now; check stores stock.

ILLUSTRATIVE · incident notes are synthetic; preset photos are drawn illustrations with pre-marked findings.

04 Four pain points, four fixes

How the week runs now

BEFORE
Progress in WhatsApp
Texts, voice notes and photos; the monthly review was the first consolidated view.

GENAI REPORT READING

AFTER
Read the same day
Activity, chainage and length extracted; a time-chainage chart with forecast finish.

BEFORE
Idle machines, missing diesel
Telematics existed but nobody compared hours, idle time and fuel across sites.

FLEET & FUEL ANALYTICS

AFTER
Flagged every morning
Variance against norm daily; under-used machines redeployed before hiring.

BEFORE
Three days per tender
RFPs ran to hundreds of pages; risky clauses surfaced late in the bid cycle.

TENDER READER

AFTER
Read in hours
Clauses, risks, estimate and a bid score before the committee meets.

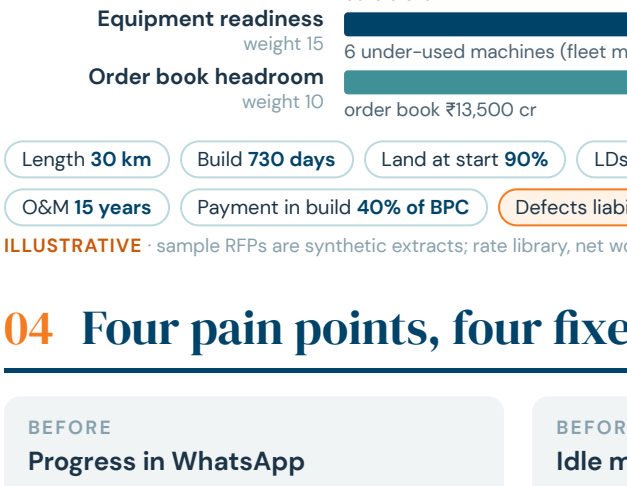
BEFORE
Safety after the incident
Paper observations, counted monthly; patterns across sites invisible.

PHOTO SAFETY CHECKS

AFTER
Site scorecard
Photo checks, sorted incidents and a score in the monthly board pack.

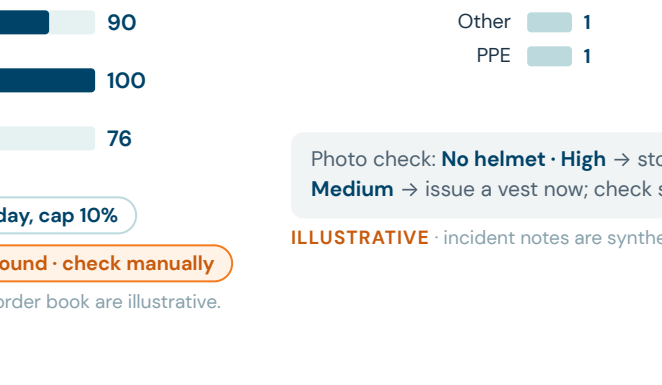
05 Closer to plan

Gap to plan at month 18 · before (light), after (navy); orange = biggest recovery



06 Less diesel lost

% of diesel issued above the norm, by site



12.0% → **3.9%** average above norm, all sites, 12 months

07 How HMC ran it

3 weeks	**Discover** What the CMD needs every Monday; how daily reports, telematics and tender checks really work.
2 weeks	**Design** Chainage as the common key; fuel norms by machine; clause risk rules.
8 weeks	**Build** Four modules, one control room: GenAI where reading is the bottleneck, rules where numbers decide.
3 pilots	**Adopt** Weekly CMD summary; plant head reviews fuel flags each morning.
All projects	**Scale** Same chainage model, full-fleet telematics, safety scorecard in the board pack.

Sections 05–06: impact charts from the case-study page; figures indicative, validated metrics available on request.

What any EPC contractor can borrow

01
Let sites report their way
Reading texts, voice notes and photos beats asking engineers to fill another form.

02
Chainage is the common key
Progress, machines, quantities and incidents all line up once tied to a kilometre.

03
Redeploy before you hire
The cheapest machine is the one already idle on another site.

Built with **OpenAI GPT (vision)** · **Whisper** · **Equipment telematics** · **Leaflet maps** · **D3** · **Power BI** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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